Pre-University Examination Questions paper collection

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Numerical Method

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Pokhara University

Everest Engineering College

Final Internal Assessment
Spring - 2025

Level: Bachelor — **F.M. 100**
Program: BE CMP(4th Semester) — **P.M. 45**
Faculty: Science & Technology — **Time: 3hrs**
Section: A/B
Subject: Numerical Methods

Attempt all the questions.

1. a) Find out at least one real root of $x^3 + x^2 - x - 1 = 0$ using Secant method to three decimal places. 7

   b) Find out all real roots of the equation $x \log_{10} x = 1.2$ using Newton Raphson method or fixed point iteration. 8

2. a) The following set of data represent the position of a car in a road at specified time, Calculate the position of the car at T = 1.75 hours. (You can use any method) 7

Time(hr)	0	0.5	1.0	1.5	2.0
Position(km)	0	0.25	1.0	2.25	4.0

OR

Find the missing value from the following table.

x	1	2	?	2.5	3.5
y	0.125	0.5275	0.7520	1.125	2.125

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b) For the following set of data, fit a curve $y = ae^{bx}e$ using least square method. 8

x_i	0.5	1.5	4.5	7.5
$f(x_i)$	2.5	3.5	6.5	9.5

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3. a) Integrate the given integral 7

$$\int_1^3 \cos x dx$$

Using trapezoidal, simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule.

b) Using two- and three-point Gauss Legendre formula. Evaluate 8

$$\int_{0.5}^{1.5} e^{x^2} dx$$

4. a) Solve the following system of linear equation using Gauss Jordan Elimination or Cholesky Method. 8

$$2w + x + 3y - z = 9$$
$$-3w + x - 2y + z = -4$$
$$w + 2x - y + 2z = 5$$
$$3w - x - y + 2z = 1$$

b) Find the largest Eigen-value and the corresponding Eigen-vector of the following square matrix using power method. 7

$$\begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

5. a) Find the value of y at x = 2.0 for the following initial value problem by using Euler method. $y' = x + y^2$; $y(1) = -0.8$, step size $h = 0.5$. 8

b) Solve $\frac{d^2y}{dx^2} + \frac{2dy}{dx} + 6y = x$, with $y(0) = 0$, $y'(0) = 1$ for $y(0.2)$ taking $h = 0.2$ by RK4 method. 7

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6. a) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \le x \le 4$, $1 \le y \le 4$, with $f = 0$ on the boundary. 8

b) A steel plate is of size 15 cm × 15 cm. If two of the sides are held at 100 °C and other two sides are held at 0 °C, what is the steady state temperature at interior points assuming a grid size of 5 cm × 5 cm? 7

7. Write short notes on *(Any Two)* 2×5

a) Errors in numerical computing

b) Cubic spline interpolation

c) Solution of differential equation

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Lumbini Engineering, Management & Science College

Final Internal Exam

Level: Bachelor
Programme: BE Computer/Electrical 4th sem
Course: Numerical Methods

Year: 2025
Full Marks: 100
Time: 3 hr

Attempt all questions.

Q>1(a) Find the root of the equation $4e^{-x}\sin x - 1 = 0$ using secant method given that the root lies between 0 and 0.5 correct to three decimal place. (8)

Q>1(b) Find the square root of 18 using NR method correct to 3 decimal place. (7)

Q>2(a) Find the root of the equation $x^3 + x^2 - 1 = 0$ correct to 6 decimal place using fixed point iteration method. (8)

Q>2(b) A third degree polynomial passes through (0,1), (1,-1), (2,-1) and (3,2). Find its value at x = 4 using forward difference interpolation. (7)

Q>3(a) Find the distance moved by a particle and its acceleration at the end of 4 seconds if the time versus velocity data is as follows: (8)

Time(t)	0	1	3	4
Velocity (v)	21	15	12	10

Q>3(b) Fit the exponential curve $y = ae^{bx}$ to the following data (7)

x	2	4	6	8	10
y	25	38	56	84	104

Q>4(a) Using Gaussian 2 point and 3 point quadrature formula, evaluate $\int (\tan^{-1} x/x)\, dx$ with lower limit 0 and upper limit 1. (8)

Q>4(b) Solve the following set of equation using LU factorization method (7)

$$x + 2y + 3z = 5$$
$$2x + 8y + 22z = 6$$
$$3x + 22y + 82z = -10$$

Q>5(a) Find the largest eigen value and corresponding eigen vector using power method (8)

$$A = \begin{bmatrix} 15 & -4 & -3 \\ -10 & 12 & -6 \\ -20 & 4 & -2 \end{bmatrix}$$

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Q>5(b) Solve the first order IVP by RK-1: $y' = x + y$, $y(0) = 0$ at $x = 1$ taking $h = 0.2$. (7)

Q>6(a) Solve the following boundary value problem by shooting method $y'' = 6x + 4$, $y(0) = 2$, $y(1) = 5$ by applying RK-2 method. (8)

Q>6(b) Solve the equation $\nabla^2 u = -81xy$, $0 < x < 1$, $0 < y < 1$ with $h = 1/3$, $u(0,y) = u(x,0) = 0$ & $u(1,y) = u(x,1) = 100$ (7)

Q> 7 Write short notes on: (any two) (5*2=10)

(i) Gauss Jacobi method
(ii) Errors in Numerical Methods
(iii) Interpolation

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GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Semester: Spring

Level: Bachelor
Programme: BE CE & SE IV
Course: **Numerical Methods**

Year : 2025
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
Attempt all the questions.

1. a) What are the different types of errors ? Solve the following equation $f(x)= xe^x-\cos x$ by Newton Raphson Method upto 4 decimal places. (2+6)

   b) Using secant method, find a positive real roots of the equation $x^3 - 2x+1=0$, correct to 4 decimal places. 7

2. a) From the given data find the lagrange polynomial and evaluate f(2.5). 8

x	1	2	4	5	7
F(x)	1	1.414	1.732	2.00	2.6

OR

Explain difference between interpolation and extrapolation ? Find the value of y(1.5) using cubic interpolation.

x	1	2	3	4
y	1	2	5	11

7

b) Use the suitable method to fit a curve $y=ae^{bx}$ for the following data.

x	-2	-1	0	1	2	3	4
y	38	6	0	-5	-1	130	300

3. a) Compute the following using Simpson's 1/3 rule for n= 8 with an accuracy to five digit. 8

$$\int_1^5 e^{-x^2}\,dx$$

OR

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Use Romberg integration to evaluate the following. 7

$$\int_4^{5.2} \log x \, dx$$

b) Evaluate the following using Gaussian three point Integration formula

$$\int_2^4 (x^4 + 1) dx$$

4 a) Solve the following system of Linear equations using partial pivoting method. 8

$x + y + z = 4,\ x + 4y + 3z = 8,\ x + 6y + 2z = 6$

b) Solve the following system of equations using Gauss Seidel method. 7

$3x + 2y + z = 10,\ 2x + 3y + 2z = 14,\ x + 2y + 3z = 14$

5. a) From the following differential equation estimate the value of y(1) using RK 4th order. 8

$dy/dx + 2x^2y = 4$ with $y(0) = 1$ take $h = 0.5$

b) Find the dominant eigen value and corresponding eigen vectors of the matrix below using Power method. 7

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

6. a) Solve the Poisson equation $\nabla^2 f = -10(x^2 + y^2 + 10)$ over the square with $0 \le x \le 3; 0 \le y \le 3$ and $f = 0$ on boundary. Use $h = 1$. 8

b) Solve the following differential equation for $y(0.4)$ using Heun's method. 7

$d^2y/dx^2 + 2dy/dx - 3y = 6x$; with $y(0) = 0$ and $y'(0) = 1$ (take $h = 0.2$).

7. Write short notes on: **(Any two)** 5*2

a) Shooting Method

b) Algorithm for Matrix factorization method

c) Schimidit method for Heat equation.

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2082/04/05			
Level	BE	Full Marks	70
Programme	BCE	Time	
Semester	IV	2 hrs	

Subject: - Numerical Methods

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ ***Attempt 70 Marks questions.***
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1. a) Derive the iterative formula for NR method. Use the **Newton-Raphson method** to find a root of the equation $x\sin(x) + \cos(x)$ near $x_0=\pi$. Perform four iterations. [2+5]

   b) Differentiate between false position method and bisection method. Find a real root of the equation $x^3-4x-9=0$ using the **Bisection method**, correct to three decimal places. [8]

2. a) The function f(x) is known only at the following unequal intervals:

x	5	7	11	13	17
f(x)	12	16	28	36	60

   Use **Newton's Divided Difference Interpolation Formula** to compute f (9). [7]

   b) The following data relates temperature x and the resistance R of a thermistor.

x(°C)	0	10	20	30	40
R(Ω)	100	80	65	50	40

   *Fit a **linear regression** model of the form: $\log_{10}R=a+bx$ using least square method. Find the estimated resistance at 25°C.* [8]

3. a) Use Romberg integration to estimate $\int_0^1 \frac{1}{1+x^2}dx$ to a tolerance of 10^{-4}. Compare the estimated solution with the true solution. [7]

   b) What are the needs of using numerical differentiation? A car's distance (in meters) after every 2 seconds is recorded:

Time(s)	0	2	4	6
Distance(m)	0	4	16	36

   Find the **speed and acceleration of the car at 4 seconds using central difference formula.** [8]

4. a) What are limitations of Naïve Gauss elimination method? Use partial pivoting method to solve the following simultaneous algebraic equations. [7]

   $x+2y+z=9$
   $2x+3y+3z=21$
   $3x+y+2z=17$

   b) Solve the system of equations using Jauss-jacobi method and **Gauss-Seidel method (up to four iterations) with the same initial guess and comments on speed of convergence.** [8]

   $10x+y+z=6$
   $x+10y+z=6$
   $x+y+10z=6$

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5. a) Solve the same equation $\frac{dy}{dx} = x + y;\ y(0) = 1$ using the **Improved Euler method** with step size h=0.2, find y (0.4). [7]

b) Use the **Runge-Kutta 4th order method** to solve: $\frac{dy}{dx} = x^2 + y; y(0) = 1$
Find y (0.2) using step size h=0.1. [8]

6. a) Consider the one-dimensional heat equation $u_t = u_{xx}$
Solve this equation for a rod of length 1, subject to the conditions: Boundary conditions:
$u(0, t)=0$, $u(1, t)=0$ and Initial condition: $u(x, 0) = 100\sin(\pi x)$ [7]

b) A square plate of size 1m x 1m is governed by the Laplace equation, $uxx+uyy=0$, representing its steady-state temperature distribution. The boundary conditions are as follows: **Top Edge:** $u(x,1)= 100°C$ **,Left Edge:** $u(0,y)= 0°C$ **,Bottom Edge:** $u(x,0)= 0°C$, **Right Edge:** $u(1,y)=0°C$. Use a square mesh with step size $h=\Delta x=\Delta y=1/3$ [8]

7. Write short notes on: (***any two***) [5*2=10]
   a) Algorithm of secant method
   b) Shooting method
   c) Lagrange Interpolation

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Madan Bhandari College of Engineering
Internal Assessment-2082

Level: Bachelor | **Course: Numerical Methods** | **Full Mark: 100**
Programme: BE | **Time: 3 hours** | **Pass Mark: 45**

Candidates are required to give their answers in their own words as far as practicable.

1. a) Find the root of the equation $f(x) = x^2 - 4x - 10$ correct to three decimal places by using False Position method. 7

   b) Estimate the root of the equation $f(x) = xe^x - cosx$ using Newton Raphson method correct to three decimal places. 8

2. a) From the data given below, find the number of students whose weight is between 60 and 70. 7

Weight in lbs	0-40	40-60	60-80	80-100	100-120
No. of students	250	120	100	70	50

   b) Using the method of least square, fit the curves $ax^2 + \frac{b}{x}$ to the following data 8

x	1	2	3	4
y	-1.52	0.96	8.88	7.66

OR

If P is pull required to lift a load W by means of a pulley, find the laws of form P=mW+C, (Where m and c are constants) using least square method for the following data:

P	12	15	21	25
W	50	70	100	120

3. a) Evaluate $\int_1^5 \frac{1}{x}\,dx$ using Gaussian Integration formula for n=3 and compare the value with the exact solution. 7

   b) Using the Romberg integration find the solution correct up to three decimal places. 8

$$I = \int_0^1 \frac{1}{1+x^2}\,dx$$

4. a) Solve the following system of equations using Gauss Jacobi's method. 7
   3x+2y+z=10, 2x+3y+2z=14, x+2y+3z=14

   b) Solve the following system of linear equations using partial pivoting method. 8
   x+y+z=4; x+4y+3z=8; x+6y+2z=6

5. a) Find the largest eigen value and the corresponding eigen vector of the matrix 7

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$ using power method.

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   b) Using the R-K 1st order method , find an approximate value of y corresponding to x=1, given that $\frac{dy}{dx} = \frac{y-x}{y+x}$ and y=1, when x=0 and h=0.02 8

6. a) Torsion on a square bar of size 15cm*15cm. If two of the sides are held at 100^0C and the other two sides are held at 0^0C.Calculate the steady state temperature at interior points. Assume a grid size of 5cm*5cm. 7

   b) Solve the poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \le x \le 4, 1 \le y \le 4$ with f=0 on the boundary. Take step size in x and y, h=k=1 8

7. Write short notes on: (Any two) 10
   a) Error in Numerical method.
   b) Ill conditioned and well-conditioned systems
   c) Cubic spline.

Best of Luck

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Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor | Semester: 4th | Year : 2025

Programme : BE Computer | Time : 3 hours | Full Marks: 100

Subject: Numerical Methods | Pass Marks: 45

Pre-Board Examination-2082 (Spring 2025)

Candidates are required to give their answer in their own words as far as possible. The figure in the margin indicate full marks

Attempt all the questions:

1. a) Write algorithm for bisection method and solve $e^{-x} - x$ correct upto 3 decimal place. [8]

   b) Derive Newton's Raphson method formula for solving non-linear equations and use this method to solve $x^3 - 4x - 9 = 0$ up to four decimal places. [7]

   OR

   Solve above question by secant method correct upto four decimal places.

2. a. Given the following data points, estimate f(0.45) using suitable backward interpolation: [8]

X	0.1	0.2	0.3	0.4	0.5
y	2.68	3.04	3.38	3.69	3.97

   b. Find f (1.6) using cubic spline method. [7]

X	1	2	3	4
Y	1	1.414	1.732	2

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3. a. Evaluate $\int_{0.2}^{1.5} e^{-x^2} dx$ by Romberg method correct up-to 3 decimal place taking n=2,4,8. [8]

   b. Compute $\int_1^3 sinx\ dx$ using Simpson's 1/3 and 3/8 rule taking suitable sub interval. [7]

4. a. Solve the following system of linear equations using the Gauss-Seidel method. [7]

$$5x + y - z = 10$$
$$2x + 8y + z = 11$$
$$-x + y + 4z = 3$$

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b. Solve the following system of linear equations using gauss elimination method: [8]

$x+2y+z=9,\ 2x-y+z=3,\ 3x+2y+3z=14$

OR

Use power method to find largest eigen value and its corresponding eigen vector

$$\begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

5. a. Solve $\frac{dy}{dx} = \frac{y}{x}$, $y(1)=2$, at $x=1.5$ using the RK-2 method. [7]

b. Solve $y'' + y = x$, $y(0)=0$, $y(1)=2$ and find $y(0.75)$ by shooting method. [8]

6. a. Solve the Laplace equation $\nabla^2 u=0$, $0\le x\le 1$, $0\le y\le 1$ with boundary conditions: $(0,y) = 0$, $u(x,0) = 0$, $u(1,y) = 100$, $u(x,1) = 100$ and $h = 1/3$. [7]

b. Solve the Poisson equation $\nabla^2 u = 2x^2y^2$ over the square region $0 \le x \le 3$ *and* $0 \le y \le 3$. given $(x,y)=0$ on boundary and $h=1$. [8]

7. Write short notes on any two: (2*5=10)

a) What is interpolation and curve fitting? Write its application and difference.

b) Why iterative method is better than direct method for solving system of linear equations? How gauss seidal method is different than Jacobi method?

c) Code of false position method.

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NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Assessment Fall/Spring 2025

Level: Bachelor | Year : 2025

Program: BE SE_Com (D_M) | Full Marks: 100

Course: : Numerical methods | Pass Marks: 45

Semester: II | Time : 2 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. (a) Solve: $\log x - \cos x = 0$ by secant method correct to four decimal places. 7

   b) Solve: $3x + \sin x - e^x = 0$ by Newton- Raphson method correct to four decimal places. 8

2. (a) From the following table, estimate the number of students who obtained the marks between 60 and 65: 8

Marks	30-40	40-50	50-60	60-70	70-80
No of Students	31	42	51	35	31

   (b) The following are data from steam table: 7

Temperature t	140	150	160	170	180
Pressure P -	3.685	4.854	6.302	8.076	10.225

Using Newton's formula, find pressure of the steam for t=145.

OR

Find the second degree least square curve to the following observations

x	0	1	2	4	
y	-1	3	9	27	

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3. a) Evaluate $\int_{0.2}^{1.5} e^{-x^2} dx$ by Romberg method. 8

b) Evaluate $\int_{-1.5}^{2.7} x^4 \sin(x) dx$ by Gaussian two and three-point formula. 7

4. a)

$$3x + 2y + 7z = 32$$
$$2x + 3y + z = 40$$
$$3x + 4y + z = 56$$

Solve by Crout's method. 8

(b) Find dominant Eigen value and corresponding Eigen vector of the matrix

$$\begin{bmatrix} 1 & 4 & 4 \\ 4 & 1 & 8 \\ 4 & 8 & 1 \end{bmatrix}$$

7

5. a) Solve: $10x + y - z = 11.19$
$-x + y + 10z = 35.61$
$x + 10y + z = 28$

By Gauss-Seidal method. 7

b) The mathematical model of an electric circuit is given by the equation $0.25\frac{d^2Q}{dt^2} + 3\frac{dQ}{dt} + 5Q = 24\sin 20t$ with Q=0 and $\frac{dQ}{dt} = 0$ at t=0. Using Range Kutta fourth order method find Q and $\frac{dQ}{dt}$ when t=0.2 8

6. (a) Solve: $\frac{dy}{dx} = \sin x + y, y(0) = 1$ for y(1) by Heun's method taking h=0.5 7

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b) Solve: $\nabla^2 u = -10(x^2 + y^2)$ over the square mesh with sides $x = 0 = y$, $x = 3 = y$ with $u = 0$ on the Boundary and $h = 1$. 8

7. Write short notes on: **(Any two)** 2×5

a) Classification of second order partial differential equations.

b) Cubic spline interpolating polynomial

c) Convergence of fixed point iterative method.

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Nepal Engineering College

Final Assessment

Level:	Bachelor	Year:	2025
Programme:	Computer	Full Marks:	100
Year/Part:	II/II	Pass Marks:	45
Subject:	Numerical Methods	Time:	3 hrs

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *The figures in the margin indicate full marks.*
- ✓ ***Attempt all the questions.***

1. a. Find the root of xlogx – sinx = 2 correct upto 4 decimal places using Bisection Method. [7]
   b. What is Convergence and rate of convergence in Numerical Computation? Solve x^3-x-11=0 correct upto 4 decimal places using fixed point iteration method. [8]
2. a. What is a cubic spline? Use cubic spline interpolation to evaluate f(1.5) from following data: [7]

x	1	2	3	4
y=f(x)	1	5	11	8

   b. Using Divided Difference Interpolation technique, evaluate y(7) from following data. [8]

x	2	3	6	9	10
y=f(x)	11	22	31	12	9

3. a. The following table gives the displacement, x (cms) of an object at various of time, t (seconds). Find the velocity and acceleration of the object at t = 1.6 sec. Using suitable interpolation method. [7]

t	1.0	1.2	1.4	1.6	1.8
x	9.0	9.5	10.2	11.0	13.2

   b. Evaluate the integral $\int_0^6 \frac{sinx}{1+x} dx$ using Simpson's 1/3 and 3/8 rules. Also comment on your answer. [8]
4. a. Solve the following system of linear equation using Gauss Elimination with Partial Pivoting. [7]

$$a + 2b + 3c - d = 10$$
$$2a + 3b - 3c - d = 1$$
$$2a - b + 2c + 3d = 7$$
$$3a + 2b - 4c + 3d = 2$$

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b. Solve the following system of linear equation using LU Factorization method. [8]

$2x + 4y - 6z = 8$

$x + 3y + z = 10$

$2x - 4y - 2z = -12$

5. a. Find the largest Eigen value and corresponding Eigen vector for following matrix. [7]

$$\begin{bmatrix} 2 & 5 & 1 \\ 5 & -2 & 3 \\ 1 & 3 & 10 \end{bmatrix}$$

b. Solve the following differential equation using RK-2 method $y'' - 4y' + 4y = e^{3x}$, $y(0)=0$, $y'(0)=-2$ at $x=2$ taking step-size $=1$. [8]

6. a. Solve the following BVP using shooting method $y'' + xy' + y = 3x^2 + 2$, $y(0) = 0$, $y(1)=1$ and $h = 0.25$ [7]

b. Solve the equation $Uxx + Uyy = 0$ over the square mesh with side $x=0=y$, $x=3=y$ with $u=0$ on boundary and mesh length=1. [8]

7. Write short notes on: (Any Two) 2*5=10

a. Errors in numerical calculations

b. Gauss Jacobi method of iteration

c. Algorithm of RK-1

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POKHARA ENGINEERING COLLEGE
ASSESSMENT EXAM

Level: Bachelor | Semester – Fall | Year : 2025
Programme: BE | Full Marks: 100
Course: Numerical Methods | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find the root of the equation $f(x) = x^3-2x-5$ with accuracy of 0.08% by using Bisection method. 7

   b) Estimate the root of the equation $f(x) = 3x+\sin x-e^x$ using Secant method. 8

2. a) Use appropriate method of interpolation to get f(3) from given table. 8

x	0	1	2	4	5	6
f(x)	1.0	14	15	5	6	19

   b) Use least square method for the following set of data to fit a parabolic curve and find f(2). 7

X	0	1	2	3	4
Y	-4	-1	4	11	20

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3 a) Evaluate the integral $I = \int_0^{\frac{\pi}{2}} \sin x dx$. Compare the result in both conditions for Simpson1/3 and 3/8 rule. 7

   b) The following data gives corresponding values of distance(D) travelled by a car at various time interval (T). 8

T	5	6	7	8	9
D	[illegible]	14.5	19.5	25.5	32

Find the velocity and acceleration at T =5 ,7.

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3. a) Using power method, find the smallest Eigen value and Eigen vector of the following matrix. $A = \begin{bmatrix} -2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ 8

b. Find the inverse of matrix using LU Decomposition method [A]= $\begin{bmatrix} 2 & -2 & 4 \\ 2 & 3 & 2 \\ -1 & 1 & 1 \end{bmatrix}$ 7

4. a) Use the Heun's method to estimate at x=2 for $dy/dx = 2Y/X$ and $y(1)= 2$. Assume $h = 0.25$. 8

b) Solve the given differential equation for y(0.25) using R-K 4th order method: 7

$10\, d^2y/dx^2 + (dy/dx)^2 + 6x = 0$ with $y(0)=1$ and $Y'(0)=0$ take $h=0.2$

5. a) Torsion on a square bar of size 15cm* 15cm. If two of the sides are held at 100° C and the other two sides are held at 0°C. Calculate the steady state temperature at interior points. Assume a grid size of 5cm * 5cm. 8

b) Solve the following system by using complete pivoting:

$$2x_1 + 7x_2 + 3x_3 = 5$$
$$3x_1 + 4x_2 + 9x_3 = 15$$
$$2x_1 + 3x_2 + x_3 = 9$$

7

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6. Write short notes on any two: 2×5

a) Error in Numerical method

b) Cubic Spline.

c) Laplacian equation

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POKHARA UNIVERSITY
FACULTY OF SCIENCE & TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2025 Spring		
Level	B.E.	F M	100
Program	BoCE & BSE	PM	45
Year/ Part	II/II	Time	3 Hrs

Subject: Numerical Methods

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks.

Attempt all the questions.

1.a) Find the real root of $e^x - 2x - 1 = 0$ using the bisection method. 7

b) Find the root of the equation $x^3 - 2x + 5 = 0$, correct to four places of decimal using fixed point iterative method. 8

2.a) Find the missing value from the following table using Lagrange's interpolation formula:

x:	1	2	4	5	6
f(x):	14	15	5	-	9

7

b) Find the polynomial f(x) and hence find f(6) from the following table, using Newton's divided difference

x	3	7	9	10
f(x)	160	120	72	63

8

3. a) The following data gives the melting point of an alloy of lead and zinc, t is the temperature in degrees Celsius and P is the percentage of lead in the alloy. Using Newton's interpolation formula, find the melting point of an alloy containing 75% lead.

P:	40	50	60	70	80
t:	184	204	226	250	276

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7

b) Growth of bacteria (N) in culture after t hours is given in the following table:

t	1	2	3	4	5
N	130	150	175	190	240

Find the equation to best fitting curve of the form $N = ab^t$ to the data and estimate N when $t = 0$.

8

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4.a) Evaluate the integral using the Gaussian 2-point and 3-point formula $\int_2^5 \frac{\sin x + e^x}{1+x}\,dx$. 7

b) Solve by Doolittle's method: $3x + 2y + 7z = 4,\ 2x + 3y + z = 5,\ 3x + 4y + z = 7$. 8

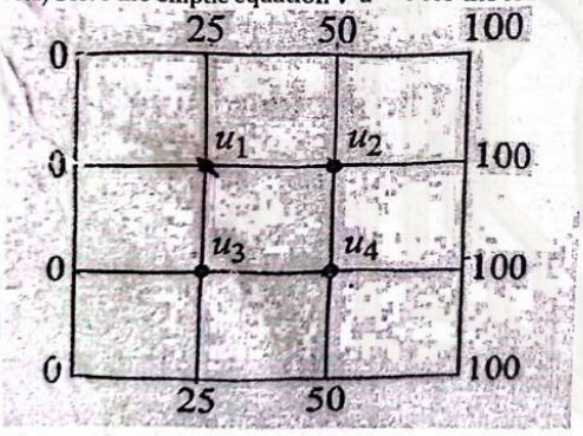
OR

Find the largest (dominant) eigenvalue and corresponding eigenvector of the matrix by the power method $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. 8

5.a) Solve the IVP by using the RK-4 method: $y' = \frac{x^2+y^2}{x+y}$, $y(0) = 1$ with step size 0.2 for $y(0.4)$. 7

b) Solve the second order differential equation using RK-2 method: $y'' + 2y' - 3y = 0,\ y(0) = 1,$ $y'(0) = 1$ with step size 0.2, for $y(0.4)$. 8

6.a) Solve the elliptic equation $\nabla^2 u = 0$ for the following square, using the Gauss-Seidel method



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OR

Solve the equation $\nabla^2 u = -81xy$ over the square domain $0 < x < 1,\ 0 < y < 1$, given that $u(0, y) = 0$, $u(x, 0) = 0, u(1, y) = 100, u(x, 1) = 100$ and mesh length $h = \frac{1}{3}$. 7

b) Solve the heat equation $u_t = u_{xx},\ 0 \le x \le 1,\ 0 \le t \le 0.1$ with boundary conditions $u(0, t) = 0 = u(1, t)$ and initial condition $u(x, 0) = \sin \pi x$, taking $h = 0.2$ and $\alpha = \frac{1}{2}$. 8

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7. Write a short note on any two: (2×5 = 10)

a) Importance and use of numerical methods in science and engineering.

b) Errors in numerical computation

c) Partial differential equations

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UNITED TECHNICAL COLLEGE

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 50
Course: Numerical Methods | Pass Marks : 23
Time : 1.5hrs.

Candidates are required to give their answers in their own words as far as practicable.

Attempt any four questions. Q.N. 7 is compulsory.

1 a Find the positive root of the equation $x^2 sinx - e^x + 2 = 0$ correct to 3 decimal places using Bisection method 7

b Evaluate the real root of $f(x) = 4\sin(x) - e^x$, using Newton's Raphson Method. The absolute error of root in consecutive iteration should be less than 0.01%. 8

2 a Estimate f (1.732) and f (2.646) from following set of data using Newton's divided difference interpolation. 7

x	-2	-1	0	1	2	3
f(x)	64	-5.5	-10	-9.5	56	366.5

b Fit an exponential function of the type $y = ae^{bx}$ to the following data. 8

x	0	0.5	1.0	1.5	2.0	2.5
y	0.10	0.45	2.15	9.15	40.35	180.75

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3 a Using Gauss 2- point and 3- point formula evaluate $\int_2^4 (x^4 + 1)dx$ 7

b Using Romberg integration method evaluate $\int_0^1 \frac{1}{1+x} dx$ 8

4 a Solve the differential equation within $0 \leq x \leq 0.5$ using RK 4th order method. 7

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$20\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 4y = 5$, y (0) = 0, y' (0) = 0. Take h=0.25.

b Find the distance moved moved by a particle and its acceleration at the end of 4 seconds, if the time versus velocity data is as follows. 8

t:	0	1	3	4
v:	21	15	12	12

5 a Find out the largest Eigen value and corresponding Eigen vector from the following square matrix 7

$$\begin{bmatrix} 2 & 2 & 1 \\ 0 & 2 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

b Solve the system of equations by applying gauss- seidel method 8

$$2x-7y-10z=-17$$
$$5x+y+3z=14$$
$$x+10y+9z=7$$

6 a Solve the poison's equation $U_{xx} + U_{yy} = 243(x^2 + y^2)$ over a square domain $0 \le x \le 1, 0 \le y \le 1$ with step size $h = \frac{1}{3}$ with $u = 100$ on the boundary. 7

b Given that $y' = x + y^2$, y(0)=1. Calculate the second approximation and hence fine the value at x=0.5 using Picard's method. 8

7 Attempt Any One (5×1=5)

a *Lagrange's Interpolation*

b *Algorithm of Simpson's 3/8 Method*

c *Algorithm of Gauss Jordan Method*

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre University Examinations

Level: Bachelor
Program: B.E. Computer
Course: **Numerical Methods**

Semester: IV_Spring

Year : 2025
F.M. : 100
P.M. : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find a real root of the equation $x^3 - 2x - 5 = 0$ with an accuracy of at least 0.0001 using Newton's method. [7]

   b) Write a difference between secant method and false position method. Using secant method find a root of the equation $3x + \sin x - e^x = 0$, correct to three decimal places. [1+7]

2. a) From the following table, find the number of students who obtain marks between 40 and 45 using suitable interpolation formula. [7]

Marks	30-40	40-50	50-60	60-70	70-80
No. of students	31	42	51	35	31

   b) The pressure and volume of a gas are related by the equation $PV^\alpha = \beta$, α and β being constants. Fit this equation to the following data. [8]

P	0.5	1.0	1.5	2.0	2.5
V	1.62	1.00	0.75	0.62	0.52

3. a) Evaluate the integral $\int_0^{\pi/2} \frac{\cos x}{\sqrt{1+\sin x}} dx$ using Gaussian quadrature formula with $n = 2$ and $n = 3$. [8]

   b) Solve the following system of linear equations using Gauss-Seidel method: [7]

   $x + 10y + 4z = 6$; $9x + 2y - 4z = 6$; $2x - 4y + 10z = -15$.

4. a) Solve the following system of equations using factorization method. [8]

   $2x + y + z = 4$; $4x + 2y + 3z = 4$; $x - y + z = 0$.

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b) Using power method, find the dominant eigen value and corresponding eigen vector of the matrix: [7]

$$\begin{pmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{pmatrix}$$

5. a) Using Runge-Kutta method of order four, solve $\frac{dy}{dx} = y + \sin x$, with $y(0) = 1$ in the interval $0 \le x \le 1$. Take $h = 0.5$. [7]

b) Solve the initial value problem
$y'' + 3y' + 2y = e^{2t}, y(0) = 1$ and $y'(0) = 1$ in the interval [0,1]

using RK-2 method with h=0.5 [8]

6. a) Solve the equation $f_{xx} + f_{yy} = 4x^2y^2$ over the square domain $0 \le x \le 3$ and $0 \le y \le 3$ with $f = 50$ on the boundary and mesh length $h = 1$ using Gauss-Seidel method correct to two decimals places. [8]

b) Find the dy/dx at x =1 using the data: [7]

x	1	1.2	1.4	1.6	1.8
y	9	9.5	10.2	11.0	12.2

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7. Attempt (Any two) [2 × 5 = 10]
   i) Using Euler's method solve the equation $y' = x^2+y$, $y(0)=1$, in the interval $0 \le x \le 1$ with $h=2.25$.
   ii) If $x=1.3506217$ is rounded off to five decimal places, calculate the absolute and relative errors.
   iii) Write short notes on ill conditioned system with an example.

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Rapti Engineering College

(Affiliated to Pokhara University)

Ghorahi Sub-Metropolitan-16 Saniambapur, Sarra, Dang

Final Internal Examination-2080

Level: Bachelor — Year : 2025

Program: Bachelor of Computer Engineering — Full Marks: 100

Semester: IV — Pass Marks: 45

Subject: *Numerical Methods* — Time: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Find the root of the equation xsinx+cosx=0 correct upto three decimal digits. Using closed end method — 8

   b) Find the square root of $\sqrt{17}$, using Newton method. — 7

2. a) Given values

x	5	7	11	13	17
f(x)	250	392	1452	2366	5202

   Calculate f(15) suitable interpolation — 7

   b) The pressure and volume of gas are related by the equation $PV^{\Upsilon} = K$, Υ and K are constants. Fit this equation to the following set of observation : — 8

P (kg/cm²)	0.5	1.0	1.5	2.0	2.5	3.0
V(liters)	1.62	1.00	0.75	0.62	0.52	0.46

3. A) Evaluate : $\int_0^6 \frac{dx}{1+x^2}$ By using — 7

   i) Trapezoidal rule

   ii) Simpson's $\frac{1}{3}$ rule

   iii) Simpson's $\frac{3}{8}$

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B) Solve the integration by Romberg integration.

$\int_0^1 \frac{dx}{1+x}$ 8

4. a) Solve the following system of linear equation by using dolittle LU decomposition method 7

$$3x_1+2x_2+x_3 = 10$$
$$2x_1+3x_2+2x_3 = 14$$
$$x_1+2x_2+3x_3 = 14$$

b) Find the largest eigen value and corresponding eigen vector by power method 8

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 3 & -5 & 2 \\ 1 & -3 & -10 \end{bmatrix}$$

5. a) Applying R-K-4 method to solve y(0.2) for the given equation $\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$ given that y=1 and $\frac{dy}{dx} = 0$ when x=0 8

b) Solve the following equation using Shooting method

$\frac{d^2y}{dx^2} = 6x$, y(1)=2 and y(2)=9 7

6 a) Solve the Poisson equation $\nabla^2 = 2x^2y^2$ over the square domain $0 \le x \le 3$ and $0 \le y \le 3$ with f=0 on the boundary and h=1. 8

b) Solve the system of linear equation using Gauss elimination method 7

$$x+2y+3z = 6$$
$$2x+3y+5z = 10$$
$$2x-y+3z = 4$$

7 Write short notes on: (any two) 2*5

a) Shooting method

b) Explain in brief the error in numerical calculations

c) ILL conditioned matrix .

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